

Unit 1, Lesson 5: Evaluating Numeric Expressions – Part 2

Benchmark

MA.7.NSO.2.1 Solve mathematical problems using multi-step order of operations with rational numbers including grouping symbols, whole-number exponents, and absolute value.

Learning Target

☐ I can evaluate numeric expressions with up to six operations.

1.5.1 Warm-Up: Growth Mindset

Famed NCAA basketball coach John Wooden once said, “Nothing will work unless you do.”

1. What do you think this quote means?
2. Describe a time where you had to work really hard to achieve something.

1.5.2 Guided Instruction: Evaluating Expressions

Two students evaluated the same expression in two different ways. Their work is shown.

Allen	Sienna
$[0.4 \times (0.4 + 4.6)^2]^3 + 24 \div (-6)$	$[0.4 \times (0.4 + 4.6)^2]^3 + 24 \div (-6)$
$[0.4 \times 5^2]^3 + 24 \div (-6)$	$0.4^3 \times (0.4 + 4.6)^{2 \times 3} + 24 \div (-6)$
$[0.4 \times 25]^3 + 24 \div (-6)$	$0.4^3 \times (0.4 + 4.6)^6 + 24 \div (-6)$
$10^3 + 24 \div (-6)$	$0.4^3 \times 5^6 + 24 \div (-6)$
$1,000 + 24 \div (-6)$	$0.064 \times 5^6 + 24 \div (-6)$
$1,000 + (-4)$	$0.064 \times 15,625 + 24 \div (-6)$
996	$1,000 + 24 \div (-6)$
	$1,000 + (-4)$
	996

- How were Allen's steps and Sienna's steps different from each other?
- Discuss with your partner whether you think one method was easier to use to evaluate the expression over the other.

1.5.3 Activity: Expression Game

1. Use a standard number cube to generate each expression. One player will roll the number cube four times each round and the other player will decide where to place each of the four values in the expression. Each round has a different way to win, stated in the first column. If in any round two players' expressions have the same value, both players lose 10 points.

Round 1 Scoring – the player with the greatest value earns 8 points.	$[(_\times_\)^2] + _\div_\$	My answer Points earned
Round 2 Scoring – the player with the least value earns 5.5 points.	$[(_\times_\)^2 + _\] \div _\$	My answer Points earned
Round 3 Scoring – the player with the greatest absolute value loses 3.7 points.	$(_\times_\)^2 + (_\div_\)$	My answer Points earned
Round 4 Scoring – the player with the absolute value closest to zero earns 11 points.	$(_\times_\)^2 + (_\div_\)$	My answer Points earned

My Game Score

Round 1	Round 2	Round 3	Round 4	My Score

2. Discuss with the other players what strategies you used to determine where to put the numbers. Summarize your discussion.

1.5.4 Try It: Using the Order of Operations

1. In the table, an expression has been evaluated, but the steps are mixed up. Order the expressions using the numbers 1 – 7 on the left. Write an explanation for each step on the right.

Step	Expression	Explain the Step
	$11 - 17.64 + 55.888$	
	$11 - (-4.2)^2 + 18.2 + (-74.088) $	
	$-6.64 + 55.888$	
1	$11 - (-4.2)^2 + 18.2 + (-4.2)^2 $	Original expression
	$11 - (-4.2)^2 + 55.888$	
	49.248	
	$11 - (-4.2)^2 + -55.888 $	

2. Evaluate each expression without using a calculator.

a. $\left(\frac{2}{3}\right)^3 \times 54 - 4^2$

b. $19 - (5 - 4^2) + \frac{2^4}{6}$

1.5.5 Activity: Class Debate

1. Aron wants to choose an expression that has a value of 1,093. He can choose between the expressions shown.

Expression A $(-3)^2 + 4 \cdot (6^2 + 11) - 7$	Expression B $(-3)^2 + 4 \cdot 6^2 + (11 - 7)$
Expression C $[(-3)^2 + 4 \cdot 6]^2 + 11 - 7$	Expression D $(-3^2 + 4 \cdot 6)^2 + (11 - 7)$

- a. Evaluate each expression. Record your answers in the table.

Expression	Work	Answer
Expression A $(-3)^2 + 4 \cdot (6^2 + 11) - 7$		
Expression B $(-3)^2 + 4 \cdot 6^2 + (11 - 7)$		
Expression C $[(-3)^2 + 4 \cdot 6]^2 + 11 - 7$		

Expression D $(-3^2 + 4 \cdot 6)^2 + (11 - 7)$		
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b. Which expression should Aron choose? _____

1.5.6 Wrap-Up: Reflection

1. Complete each statement based on your learning today.

a. A barrier to my learning was ...

b. I will try to overcome this by ...

